

Sydney Girls High School

2014

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 3 – 6

10 Marks

- Attempt Questions 1 – 10
- Answer on the Multiple Choice answer sheet provided
- Allow about 15 minutes for this section

Section II Pages 7 – 13

60 Marks

- Attempt Questions 11 – 14
- Answer on the blank paper provided
- Begin a new page for each question
- Allow about 1 hour and 45 minutes for this section

Name:

Teacher:

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format or the content of the 2014 HSC Examination Paper in this subject.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

(1) What are the values of p such that $\frac{p+1}{p} \leq 1$?

- (A) $p > 0$
- (B) $p < 0$
- (C) $p \leq 0$
- (D) $-1 \leq p \leq 0$

(2) The expression $\tan\left(\frac{\pi}{4} + x\right)$ can also be expressed as:

- (A) $\frac{\cos x + \sin x}{\cos x - \sin x}$
- (B) $\frac{\cos x - \sin x}{\cos x + \sin x}$
- (C) $\frac{\sec^2 x}{1 - \tan^2 x}$
- (D) $\frac{\sin x + \cos x}{\sin x - \cos x}$

(3) The acute angle (to the nearest degree) between the lines $x - y = 2$ and $2x + y = 1$ is:

- (A) 18°
- (B) 27°
- (C) 45°
- (D) 72°

(4) Two of the roots of the polynomial $4x^3 + 8x^2 + kx - 18 = 0$ are equal in magnitude but opposite in sign. Find the value of k .

- (A) $k = -2$
- (B) $k = 2$
- (C) $k = -9$
- (D) $k = 9$

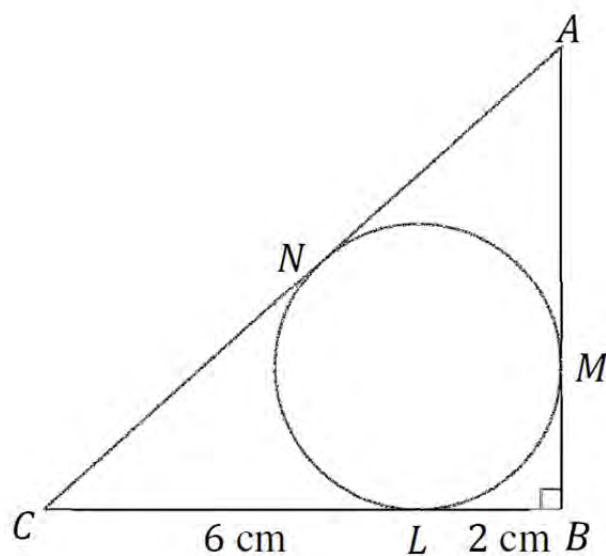
(5) $y = f(x)$ is a linear function with slope $\frac{1}{3}$, find the slope of $y = f^{-1}(x)$.

- (A) 3
- (B) $\frac{1}{3}$
- (C) -3
- (D) $-\frac{1}{3}$

(6) In the diagram, AC is a tangent to the circle at the point N , AB is a tangent to the circle at the point M and BC is a tangent to the circle at the point L .

Find the exact length of AM if $CL = 6$ cm and $BL = 2$ cm .

- (A) 3 cm
- (B) 4 cm
- (C) 5 cm
- (D) 6 cm



(7) Find $\int \frac{dx}{1+4x^2}$

(A) $\frac{1}{2} \tan^{-1} 2x + C$

(B) $2 \tan^{-1} 2x + C$

(C) $2 \tan^{-1} \frac{x}{2} + C$

(D) $\frac{1}{2} \tan^{-1} \frac{x}{2} + C$

(8) Evaluate $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$.

(A) -10

(B) -5

(C) 5

(D) 10

(9) Using $u = x^2 + 1$, the value that is equal to $\int_0^1 3x(x^2 + 1)^5 dx$ is:

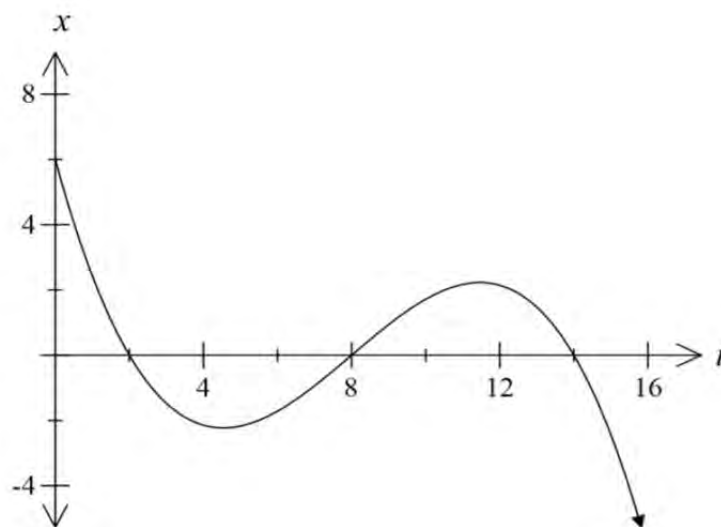
(A) $\frac{1}{4}$

(B) $\frac{16}{3}$

(C) $\frac{63}{4}$

(D) 32

- (10) The displacement, x metres, from the origin of a particle moving in a straight line at any time (t seconds) is shown in the graph.



When was the particle at rest?

- (A) $t = 4.5$ and $t = 11.5$
- (B) $t = 0$
- (C) $t = 2$, $t = 8$ and $t = 14$
- (D) $t = 8$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer on the blank paper provided. Begin a new page for each question

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11

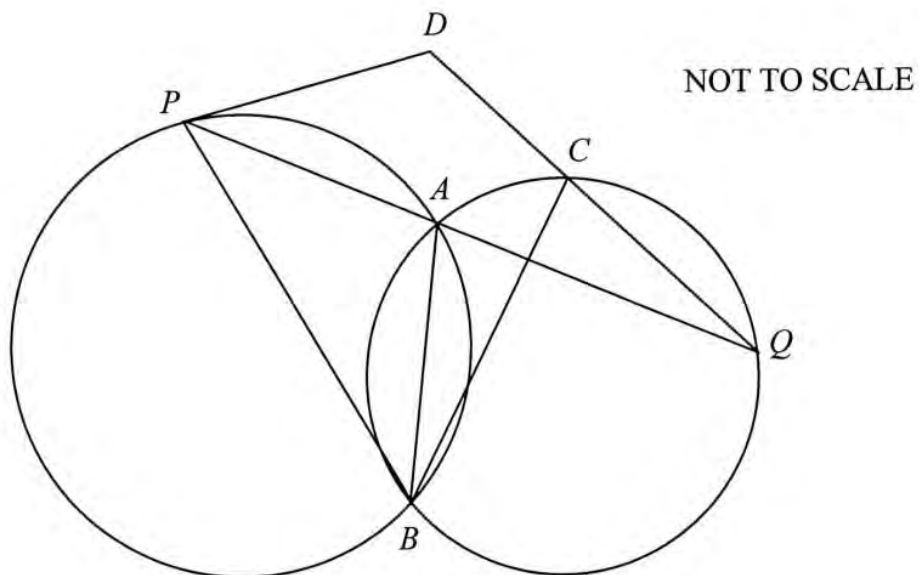
(15 Marks)

- (a) Evaluate $\int_0^{\frac{\pi}{6}} \sec 2x \tan 2x \, dx$. [2]
- (b) $A(-3, 7)$ and $B(4, -2)$ are two points. Find the coordinates of the point $P(x, y)$ which divides the interval AB internally in the ratio $3:2$. [2]
- (c) The equation $2x^3 - 6x + 1 = 0$ has roots α , β and γ . Evaluate:
- i) $\alpha + \beta + \gamma$. [1]
- ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. [2]
- (d) i) Find the domain and range of the function $f(x) = 2\cos^{-1}(1-x)$. [2]
- ii) Sketch the graph of the curve $y = 2\cos^{-1}(1-x)$ showing clearly the coordinates of the endpoints. [2]

Question 11 continues on the next page

Question 11 (Continued)

(e)



Two circles intersect at A and B. P is a point on the first circle and Q is a point on the second circle such that PAQ is a straight line. C is a point on the second circle. The line QC produced and the tangent to the first circle at P meet at D.

- i) Copy the diagram.
- ii) Give a reason why $\angle DPA = \angle PBA$. [1]
- iii) Give a reason why $\angle CQA = \angle CBA$. [1]
- iv) Hence show that BCDP is a cyclic quadrilateral. [2]

End of Question 11

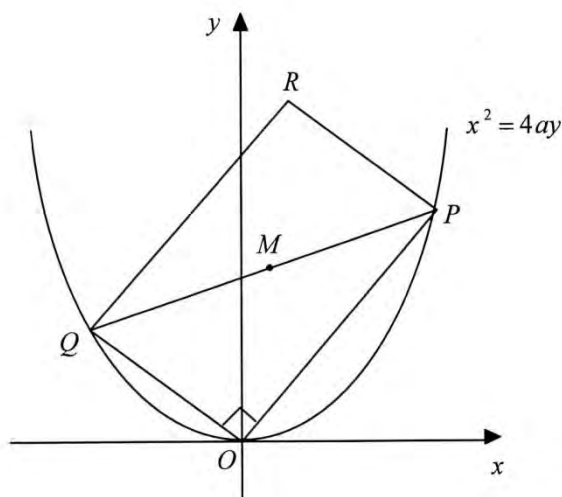
Question 12 (Begin a New Page)

(15 Marks)

- (a) Use the method of Mathematical Induction to show that $5^n + 12n - 1$ is divisible by 16, for all positive integers $n \geq 1$.

[3]

(b)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points which move on the parabola $x^2 = 4ay$ such that $\angle POQ = 90^\circ$, where O is the origin.

$M = \left(a(p+q), \frac{1}{2}a(p^2 + q^2) \right)$ is the midpoint of PQ . R is the point such that $OPRQ$ is a rectangle.

- i) Show that $pq = -4$. [1]
- ii) Show that R has coordinates $(2a(p+q), a(p^2 + q^2))$. [1]
- iii) Find the equation of the locus of R . [2]

Question 12 continues on the next page

Question 12 (Continued)

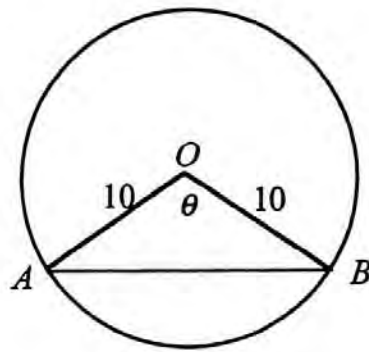
(c)

i) Show that the equation $e^x + x = 0$ has a real root α such that $-1 < \alpha < 0$. [2]

ii) If a is taken as an initial approximation to this real root α , use Newton's [2]
method to show that the next approximation a_1 is given by $a_1 = \frac{(a-1)e^a}{e^a + 1}$.

Hence if the initial approximation is taken as $a = -0.5$, find the next approximation for α correct to two decimal places.

(d)



The chord AB of a circle of radius 10 cm subtends an angle θ radians at the centre O of the circle.

i) Show that the perimeter P cm of the minor segment cut off by the [2]
chord AB is given by $P = 10\theta + 20 \sin \frac{\theta}{2}$.

ii) If θ is increasing at a rate of 0.02 radians per second, find the rate at [2]
which P is increasing when $\theta = \frac{2\pi}{3}$.

End of Question 12

Question 13 (Begin a New Page)**(15 Marks)**

(a) Evaluate $\int_1^{49} \frac{1}{4(x + \sqrt{x})} dx$ using the substitution $u^2 = x$, $u > 0$. [4]

Give the answer in simplest exact form.

(b) Newton's Law of Cooling states that the rate of change in the temperature, T , of a body is proportional to the difference between the temperature of the body and the surrounding temperature, P .

i) If A and k are constants, show that the equation $T = P + Ae^{kt}$ satisfies Newton's Law of Cooling. [2]

ii) A cup of tea with temperature of 100°C is too hot to drink. Two minutes later, the temperature has dropped to 93°C . If the surrounding temperature is 23°C , calculate the value of A and k (correct to 3 significant figures). [2]

iii) The tea will be drinkable when the temperature has dropped to 80°C . [1]
How long in minutes will this take?

(c) A particle's motion is defined by the equation $v^2 = 12 + 4x - x^2$, where x is its displacement from the origin in metres and v its velocity in ms^{-1} . Initially, the particle is 6 metres to the right of the origin.

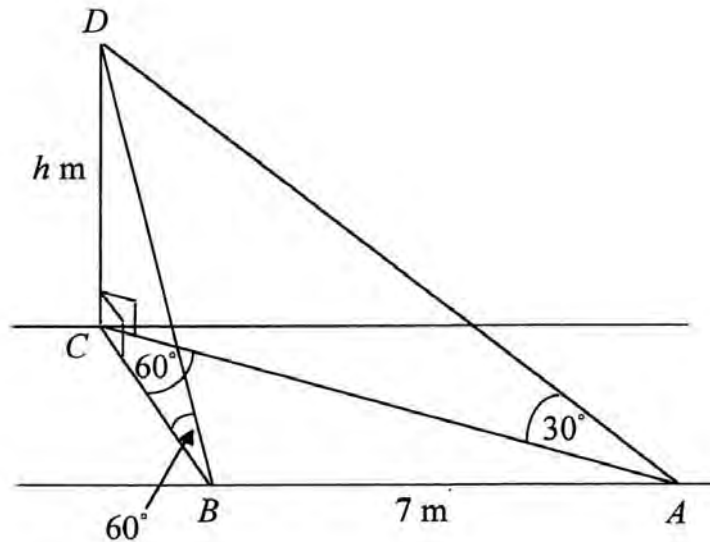
i) Show that the particle is moving in Simple Harmonic Motion. [1]

ii) Find the centre, the period and the amplitude of the motion. [3]

iii) The displacement of the particle at any time t is given by the equation $x = a \sin(nt + \theta) + b$. Find the values of θ and b , given $0 \leq \theta \leq 2\pi$. [2]

End of Question 13

(a)



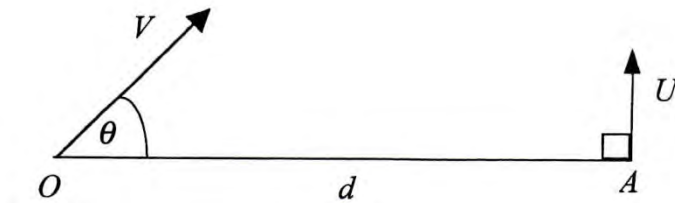
A footpath on horizontal ground has two parallel edges. CD is a vertical flagpole of height h metres which stands with its base C on one edge of the footpath. A and B are two points on the other edge of the footpath such that $AB = 7$ m and $\angle ACB = 60^\circ$. From A and B the angles of elevation of the top D of the flagpole are 30° and 60° respectively.

- i) Find the exact height of the flagpole. [3]
- ii) Find the exact width of the footpath. [2]

Question 14 continues on the next page

Question 14 (Continued)

(b)



O and A are two points d metres apart on horizontal ground. A rocket is projected from O with speed $V \text{ ms}^{-1}$ at an angle θ above the horizontal where $0 < \theta < \frac{\pi}{2}$. At the same instant, another rocket is projected vertically from A with speed $U \text{ ms}^{-1}$.

The two rockets move in the same vertical plane under gravity where the acceleration due to gravity is $g \text{ ms}^{-2}$.

After time t seconds, the rocket from O has horizontal and vertical displacements x metres and y metres respectively from O , while the rocket from A has vertical displacement Y metres from A . The two rockets collide after T seconds.

i) Derive the expressions for x , y and Y in terms of V , θ , U , t and g . [3]

ii) Show that $d = VT \cos \theta$ and $U = V \sin \theta$. [2]

iii) Show that $V > U$. [1]

iv) Show that the two rockets are the same distance above ground level at all times. [1]

v) Show that $T = \frac{d}{\sqrt{V^2 - U^2}}$. [2]

vi) If the two rockets collide at the highest points of their flights, show that [1]

$$d = \frac{U\sqrt{V^2 - U^2}}{g}.$$

End of Exam

Ext 1.



Sydney Girls High School

Mathematics Faculty

Multiple Choice Answer Sheet

Mathematics

2014 EXT 1 THSC

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☒ C ☐ D ☐
2. A ☒ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☒
4. A ☐ B ☐ C ☒ D ☐
5. A ☒ B ☐ C ☐ D ☐
6. A ☐ B ☒ C ☐ D ☐
7. A ☒ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☒ D ☐
9. A ☐ B ☐ C ☒ D ☐
10. A ☒ B ☐ C ☐ D ☐

Multiple choice.

$$\begin{aligned} \textcircled{1} \quad \frac{p+1}{p} &\leq 1 \\ p^2 \left(\frac{p+1}{p} \right) &\leq p^2 \quad (p \neq 0) \\ p(p+1) &\leq p^2 \\ p^2 + p &\leq p^2 \\ p &\leq 0 \quad p \neq 0. \end{aligned}$$

$$\therefore \underline{p < 0} \quad \textcircled{B}.$$

$$\begin{aligned} \textcircled{2} \quad \tan \left(\frac{\pi}{4} + x \right) &= \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x} \\ &= \frac{1 + \frac{\sin x}{\cos x}}{1 - \frac{\sin x}{\cos x}} \\ &= \frac{\cos x + \sin x}{\cos x} \div \frac{\cos x - \sin x}{\cos x} \\ &= \frac{\cos x + \sin x}{\cos x} \times \frac{\cos x}{\cos x - \sin x} \\ &= \frac{\cos x + \sin x}{\cos x - \sin x} \quad \textcircled{A} \end{aligned}$$

Multiple choice.

$$\textcircled{3} \quad x - y = 2 \Rightarrow y = x - 2 \quad \therefore m_1 = 1$$

$$2x + y = 1 \Rightarrow y = 1 - 2x \quad \therefore m_2 = -2.$$

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{1 - (-2)}{1 + (1)(-2)} \right| \\ &= \left| \frac{3}{-1} \right| \end{aligned}$$

$$\therefore \tan \theta = 3$$

$$\underline{\theta = 72^\circ} \quad \textcircled{D}.$$

$$\textcircled{4} \quad 4x^3 + 8x^2 + kx - 18 = 0.$$

$$\alpha + (-\alpha) + \beta = -\frac{8}{4} = -2$$

$$\therefore \beta = -2.$$

$$\alpha(-\alpha)\beta = \frac{18}{4}$$

$$-\alpha^2(-2) = \frac{9}{2}$$

$$\alpha^2 = \frac{9}{4}$$

$$\begin{aligned} \alpha(-\alpha) + \alpha\beta + (-\alpha\beta) &= \frac{k}{4} \\ -\alpha^2 + \cancel{\alpha\beta} - \cancel{\alpha\beta} &= \frac{k}{4} \end{aligned}$$

$$-\alpha^2 = \frac{k}{4}$$

$$-\frac{9}{4} = \frac{k}{4} \quad \therefore \underline{k = -9} \quad \textcircled{C}$$

Multiple choice .

⑤ $f(x): y = mx + b$
 $y = \frac{1}{3}x + b$

$f^{-1}(x): x = \frac{1}{3}y + b$

$3x = y + 3b$

$y = 3x - 3b$

$\therefore m = 3$

(A)

⑥ $CL = CN = 6 \text{ cm}$

$LB = BM = 2 \text{ cm}$

$AN = AM = x \text{ cm}$

$(x+2)^2 + 8^2 = (x+6)^2$

$x^2 + 4x + 4 + 64 = x^2 + 12x + 36$

$4x + 68 = 12x + 36$

$32 = 8x$

$x = 4$

$\therefore AM = 4 \text{ cm}$

(B)

⑦ $\int \frac{dx}{1+4x^2}$

$= \int \frac{dx}{4(\frac{1}{4} + x^2)}$

$= \frac{1}{4} \cdot \frac{1}{\frac{1}{2}} \tan^{-1} \frac{x}{\frac{1}{2}}$

$= \frac{1}{4} \cdot 2 \tan^{-1} 2x$

$= \frac{1}{2} \tan^{-1} 2x + C$

(A)

Multiple choice .

⑧ $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$

$= \lim_{x \rightarrow 0} \frac{5x(1 - 2\sin^2 x)}{\sin x}$

$= \lim_{x \rightarrow 0} \frac{5x}{\sin x} - \frac{10x \sin^2 x}{\sin x}$

$= \lim_{x \rightarrow 0} \frac{5x}{x} - 10x^2_0 \quad (\sin x \div x, \text{ as } x \rightarrow 0)$

$= 5$

(C)

⑨ $u = x^2 + 1 \quad x = 1, u = 2$

$du = 2x dx \quad x = 0, u = 1$

$\int_0^1 3x(x^2+1)^5 dx$

$= \int_1^2 3x \cdot u^5 \cdot \frac{du}{2x}$

$= \frac{3}{2} \int_1^2 u^5 du$

$= \frac{3}{2 \times 6} [u^6]_1^2$

$= \frac{1}{4} \times [2^6 - 1]$

$= \frac{63}{4}$

(C)

⑩ (A)

Question 11 - 15 marks - Ext I Mathematics - 2014 - Trials

$$\begin{aligned} \text{a) } \int_0^{\pi/6} \sec 2x \tan 2x \, dx &= \frac{1}{2} \sec 2x \Big|_0^{\pi/6} \\ &= \frac{1}{2} \sec \frac{\pi}{3} - \frac{1}{2} \sec 0 \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \quad (2 \text{ marks}) \end{aligned}$$

- Overall this question was done poorly. Many solved it by substitution

$$\begin{aligned} \text{b) } m:n & \quad \frac{x_2 m + x_1 n}{m+n}, \frac{y_2 m + y_1 n}{m+n} \\ 3:2 & \\ A(-3, 7) & \quad \frac{4(3) + (-3)(2)}{3+2}, \frac{(-2)(3) + (7)(2)}{3+2} \\ x_1, y_1 & \\ B(4, -2) & \quad \frac{12-6}{5}, \frac{-6+14}{5} \\ x_2, y_2 & \\ \therefore P & \left(\frac{6}{5}, \frac{8}{5} \right) \text{ or } \left(1\frac{1}{5}, 1\frac{3}{5} \right) \quad (2 \text{ marks}) \end{aligned}$$

- Overall done very well. Most errors were due to carelessness.

$$\text{c) } 2x^3 + 0x^2 - 6x + 1 = 0$$

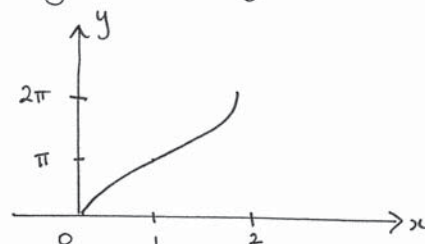
$$\begin{aligned} \text{i) } \alpha + \beta + \gamma &= -\frac{b}{a} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{c}{a} \div -\frac{d}{a} \\ &= -\frac{6}{2} \times \frac{2}{-1} \\ &= 6 \end{aligned}$$

- Overall done very well, except for some who failed to recognise x^2
 $2x^3 + 0x^2 - 6x + 1 = 0$
 was missing!

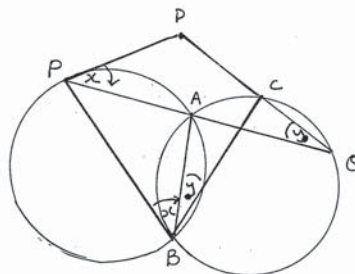
$$\text{d) i) Domain: } 0 \leq x \leq 2$$

$$\text{Range: } 0 \leq y \leq 2\pi$$



Overall domain & range was found very well. Sketches of the graph varied. Most sketched correctly.

e)



Parts ii) & iii) were completed extremely well. iv) Most completed question well, some proved it by the exterior $\angle$ equals interior opposite $\angle$ of cyclic quad.

$$\text{ii) } \angle DPA = \angle PBA \text{ (angle in the alternate segment)} = x^\circ$$

$$\text{iii) } \angle CQA = \angle CBA \text{ (angles standing on the same arc)} = y^\circ$$

$$\text{iv) } \angle D = 180 - (x+y) \text{ (angle sum of } \triangle POQ)$$

$$\begin{aligned} \therefore \angle D + \angle PBC &= 180 - (x+y) + x+y \\ &= 180^\circ \end{aligned}$$

$\therefore$ opposite $\angle$'s of quadrilateral BCDP are supplementary, hence, BCDP is a cyclic quadrilateral.

Ext 1 2014 Trial

$$12a) \quad n=1$$

$$= 5+12-1$$

$$= 16$$

Divisible by 16

$$5^k + 12k - 1 = 16p$$

Prove true for $n=k+1$

$$5^{k+1} + 12(k+1) - 1$$

$$= 5 \cdot 5^k + 12k + 12 - 1$$

$$= 5(16p - 12k + 1) + 12k + 12 - 1$$

$$= 80p - 60k + 5 + 12k + 12 - 1$$

$$= 80p - 48k + 16$$

$$= 16(5p - 3k + 1)$$

$$= 16q$$

*Most students did this question well, but a few didn't do the last main section well.

$$b(i) \quad m_{aq} \times m_{op} = -1$$

$$\frac{aq^2 - 0}{2aq} \times \frac{ap^2 - 0}{2ap} = -1$$

$$\frac{q \times p}{4} = -1$$

$$pq = -4$$

$$b(ii)$$

$$a(p+q), \frac{1}{2}a(p^2+q^2), \frac{x+0}{2}, \frac{y+0}{2}$$

$$\frac{x}{2} = a(p+q), \frac{y}{2} = \frac{1}{2}a(p^2+q^2)$$

$$x = 2a(p+q), y = a(p^2+q^2)$$

*Many students did more than a page of working for a one mark question

$$b(iii)$$

$$x = 2a(p+q)$$

$$y = a(p^2+q^2)$$

$$p+q = \frac{x}{2a}$$

$$y = a\left[\left(\frac{x}{2a}\right)^2 - 2pq\right]$$

$$= a\left[\left(\frac{x}{2a}\right)^2 + 8\right]$$

$$y = \frac{x^2}{4a} + 8a$$

$$x^2 = 4a(y - 8a)$$

* This question was done better than part i) and ii)

$$c)i)$$

$$f(x) = e^x + x$$

$$f(0) = e^0 + 0$$

$$= e > 0$$

$$f(-1) = e^{-1} - 1$$

$$= -0.63 < 0$$

since $f(0) > 0$ and $f(-1) < 0$: there is a root $-1 < x < 0$

*The setting out for this part was very poor.

$$ii) \quad x_1 = x - \frac{f(x)}{f'(x)}$$

$$a_1 = a - \frac{e^a + a}{e^a + 1}$$

$$= \frac{a(e^a + 1) - e^a - a}{e^a + 1}$$

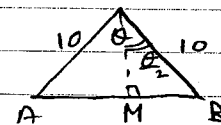
$$= \frac{ae^a e^a + 1 - e^a - a}{e^a + 1}$$

$$= \frac{e^a(e^a - 1)}{e^a + 1}$$

$$= -0.57$$

*Please remember for a "show" question you need to show all the steps.

d)i)



$$\sin \frac{\theta}{2} = \frac{MB}{10}$$

$$MB = 10 \sin \frac{\theta}{2}$$

$$AB = 2MB$$

$$AB = 20 \sin \frac{\theta}{2}$$

$$P = r\theta + 20 \sin \frac{\theta}{2}$$

$$P = 10\theta + 20 \sin \frac{\theta}{2}$$

*This was the simplest way of doing this question. some students used a harder method and didn't show the working properly.

$$ii) \quad \frac{d\theta}{dt} = 0.02$$

$$\frac{dP}{dt} = \frac{dP}{d\theta} \cdot \frac{d\theta}{dt}$$

$$= (10 + 20 \times \frac{1}{2} \times \cos \frac{\theta}{2}) \times 0.02$$

$$= (10 + 10 \cos \frac{\theta}{2}) \times 0.02$$

$$= 15 \times 0.02$$

$$= 0.3 \text{ cm/s}$$

*some students didn't get the $\cos \frac{\theta}{2}$ correctly

Question 13:

a) $I = \int_1^{49} \frac{1}{4(x+\sqrt{x})} dx$

$$\left[\begin{array}{l|l} u^2 = x, u > 0 & \\ \hline 2u = \frac{dx}{du} & x=1 \Rightarrow u=1 \\ \hline \therefore dx = 2u du & x=49 \Rightarrow u=7 \end{array} \right] \checkmark$$

$$\therefore I = \int_1^7 \frac{1}{4(u^2+u)} 2u du$$

$$= \int_1^7 \frac{1}{2(u+1)} du$$

* Students who did not factorise at this point, lost unnecessary marks.

$$= \frac{1}{2} \left[\ln(u+1) \right]_1^7 \checkmark$$

$$= \frac{1}{2} (\ln 8 - \ln 2)$$

$$= \frac{1}{2} \ln 4$$

$$= \ln 2 \text{ (exact answer)} \checkmark$$

* Carry on error from integration was accepted.

* Alternate solutions accepted.

Question 13

b) i)

Newton's Law is $\frac{dT}{dt} = k(T-P)$ ✓

$$\text{If } T = P + Ae^{kt}$$

$$\text{then } \frac{dT}{dt} = k \times Ae^{kt}$$

$$\therefore \frac{dT}{dt} = k(T-P) \checkmark$$

ii) When $T=100$, $P=23$, $t=0$:

$$100 = 23 + Ae^0$$

$$\therefore A = 77 \checkmark$$

When $t=2$, $T=93$

$$93 = 23 + 77e^{k \times 2}$$

$$70 = 77e^{2k}$$

$$\frac{70}{77} = e^{2k}$$

* Rounding-off correct to 3. sig. figs needs ✓ to be REVISED by MOST students.

$$\therefore k = \frac{1}{2} \ln \frac{70}{77} \doteq -0.0477 \text{ (3 sig. figs)}$$

$$\text{iii) } 80 = 23 + 77e^{-0.0477 \times t}$$

$$\frac{57}{77} = e^{-0.0477t}$$

$$\therefore t = \frac{1}{-0.0477} \ln \frac{57}{77}$$

$$t = 6.31106047 \text{ min}$$

$$\text{or } t = 7 \text{ min}$$

* Rounding to 6 min is incorrect (without previous work/approximation).

Question 13

c) $v^2 = 12 + 4x - x^2$

i) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$
 $= \frac{d}{dx} \left(6 + 2x - \frac{x^2}{2} \right)$

$= 2 - x$

$= -1(x-2) \equiv -n^2(x-b)$

$\therefore$ particle moves in SHM. ✓

ii) Centre of motion is $x=2$ (where $a=0$). ✓

$n=1$ so period $T = \frac{2\pi}{n} = 2\pi$ ✓

Extremes of motion when $v=0$:

$12 + 4x - x^2 = 0$

$(6-x)(2+x) = 0$

$\therefore x = -2$ and $x = 6$

$\therefore$ amplitude of motion is 4. ✓

iii) $a=4, n=1, b=2$

$\therefore x = 4 \sin(t+\theta) + 2$ ✓

when $t=0, x=6$:

$6 = 4 \sin \theta + 2$

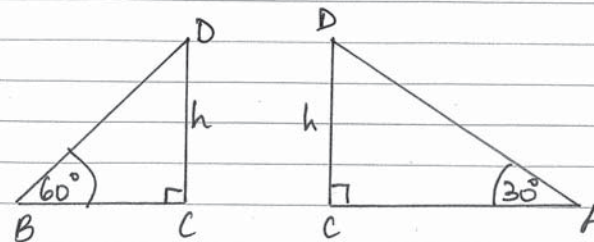
$4 = 4 \sin \theta$

$\therefore \theta = \frac{\pi}{2}$ ✓

$\therefore x = 4 \sin \left(t + \frac{\pi}{2} \right) + 2$

Question 14:

a)



In $\triangle BCD$: $\frac{h}{BC} = \tan 60^\circ$

$\therefore BC = \frac{h}{\tan 60^\circ} = h \cot 60^\circ = \frac{h}{\sqrt{3}}$ ✓

In $\triangle ACD$: $\frac{h}{AC} = \tan 30^\circ$

$\therefore AC = \frac{h}{\tan 30^\circ} = h \cot 30^\circ = \sqrt{3}h$ ✓

Using the cosine rule in $\triangle ABC$:

$AB^2 = BC^2 + AC^2 - 2(BC)(AC) \cos 60^\circ$

$7^2 = \frac{h^2}{3} + 3h^2 - 2 \times \frac{h}{\sqrt{3}} \times \sqrt{3}h \times \frac{1}{2}$

$49 = h^2 \left(\frac{1}{3} + 3 - 1 \right)$

$49 = \frac{7}{3} h^2$

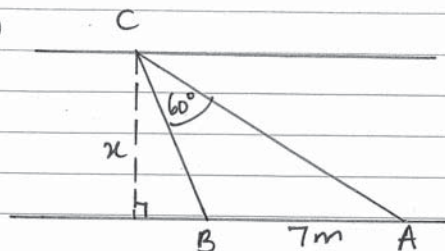
$h^2 = \frac{3 \times 49}{7}$

$h = \sqrt{21} \text{ m } (h > 0)$ ✓

* Correct answers, not in exact form, were awarded a mark.

Question 14

a) ii)



Let the width of the footpath be x metres.

$$\text{Area } \triangle ABC = \frac{1}{2} ab \sin C$$

$$\frac{1}{2} \times 7 \times x = \frac{1}{2} \times BC \times AC \times \sin 60^\circ$$

$$7x = \frac{h}{\sqrt{3}} \times \sqrt{3} h \times \frac{\sqrt{3}}{2}$$

$$x = \frac{\sqrt{3}}{14} h^2$$

$$x = \frac{21\sqrt{3}}{14}$$

$$x = \frac{3\sqrt{3}}{2} \text{ m}$$

* Many students thought that $\triangle ABC$ is right-angled. This was awarded ZERO marks. The problem was over-simplified.

Question 14:

b) i) Rocket from point O:

horizontal motion:

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\text{when } t=0, \dot{x} = V \cos \theta$$

$$\therefore \dot{x} = V \cos \theta$$

$$x = V \cos \theta t + C_1$$

$$\text{when } t=0, V=0$$

$$\therefore x = V \cos \theta t \quad \checkmark$$

vertical motion:

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c$$

$$\text{when } t=0, \dot{y} = V \sin \theta$$

$$\therefore \dot{y} = V \sin \theta - gt$$

$$y = V \sin \theta t - \frac{gt^2}{2} + C_2$$

$$\text{when } t=0, V=0 \therefore C_2 = 0$$

$$\therefore y = V \sin \theta t - \frac{1}{2} gt^2 \quad \checkmark$$

* Students who did not DERIVE the equations lost one mark.

Question 14:

bi) Rocket from point A:

$$\ddot{Y} = -g$$

$$\dot{Y} = -gt + c$$

$$\text{when } t=0, \dot{Y} = U$$

$$\therefore \dot{Y} = U - gt$$

$$Y = Ut - \frac{gt^2}{2} + c_1$$

$$\text{when } t=0, Y=0 \therefore c_1=0$$

$$\therefore Y = Ut - \frac{gt^2}{2} \quad \checkmark$$

ii) When the rockets collide at time T ,
they must be vertically above A with the same height.

$$x = Vt \cos \theta$$

$$\text{when } t=T, x=d:$$

$$\therefore d = VT \cos \theta \quad \checkmark$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2$$

$$\text{when } t=T, y=Y:$$

$$\therefore UT - \frac{gt^2}{2} = VT \sin \theta - \frac{gt^2}{2}$$

$$UT = VT \sin \theta$$

$$\therefore U = V \sin \theta \quad \checkmark$$

Question 14

b) iii) $U = V \sin \theta$

$$\text{since } 0 < \theta < \frac{\pi}{2}$$

$$\text{then } 0 < \sin \theta < 1 \quad \checkmark$$

$$\therefore V > V \sin \theta$$

$$\therefore V > U$$

$$\begin{aligned} \text{iv) } Y &= Ut - \frac{1}{2}gt^2 \\ &= (V \sin \theta)t - \frac{1}{2}gt^2 \quad \checkmark \\ &= y \end{aligned}$$

Hence the rockets are always at the same height above ground level.

$$v) V \cos \theta = \frac{d}{T} \quad (\text{from ii})$$

$$V \sin \theta = U$$

$$\therefore V^2 (\cos^2 \theta + \sin^2 \theta) = \frac{d^2}{T^2} + U^2 \quad \checkmark$$

$$V^2 = \frac{d^2}{T^2} + U^2$$

$$\therefore V^2 - U^2 = \frac{d^2}{T^2}$$

$$T^2 = \frac{d^2}{V^2 - U^2} \quad \checkmark$$

$$\therefore T = \frac{d}{\sqrt{V^2 - U^2}}, (T > 0)$$

* alternate solutions accepted.

Question 14:

b) vi) At the highest point of flight of the rocket from A:

$$\dot{y} = 0$$

$$V - gt = 0$$

$$V = gt$$

$$\therefore t = \frac{V}{g}$$

$\therefore$ Rockets collide at highest point if $T = \frac{V}{g}$.

Then $d = T\sqrt{V^2 - U^2}$

$$\therefore d = \frac{U\sqrt{V^2 - U^2}}{g}, \text{ as required. } \checkmark$$